

[illegible]

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drift per km, against
a 10 % ceiling



SETU — NAVIGATION WITHOUT GNSS



When GPS disappears, today's apps freeze, jump, or call the turn too late. **SETU keeps your position moving correctly on the map, using nothing but the sensors already inside the phone.**

Detailed explanation of the proposed solution

An Android app and a small on-device engine. No dongle, no extra hardware, no internet.

The phone hears the wheels.
Spinning wheels shake the car, and the shake speeds up as you do. The app reads that pitch, so it measures speed instead of guessing it from acceleration — the way a mechanic listens to an engine.

Every turn re-checks the speed.
In a bend, the sideways push divided by how sharply you are turning gives the exact speed — free calibration several times a minute, and it works on a leaning two-wheeler too.

The road becomes the ruler.
The phone knows how much it has turned; the offline map knows the shape of every road. Line the two up and your position falls into place with no satellites at all.

Innovation and uniqueness of the solution

N1 Speed read from frequency

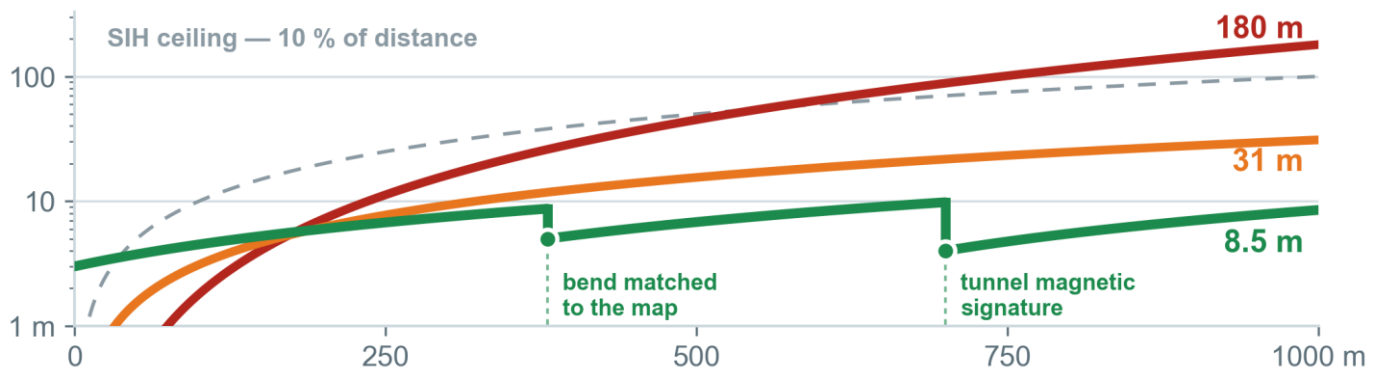
N2 Two-wheeler lean physics

N3 Scale-free map registration

N4 CAN wheel speeds as teacher

N5 Learned σ and Q in one filter

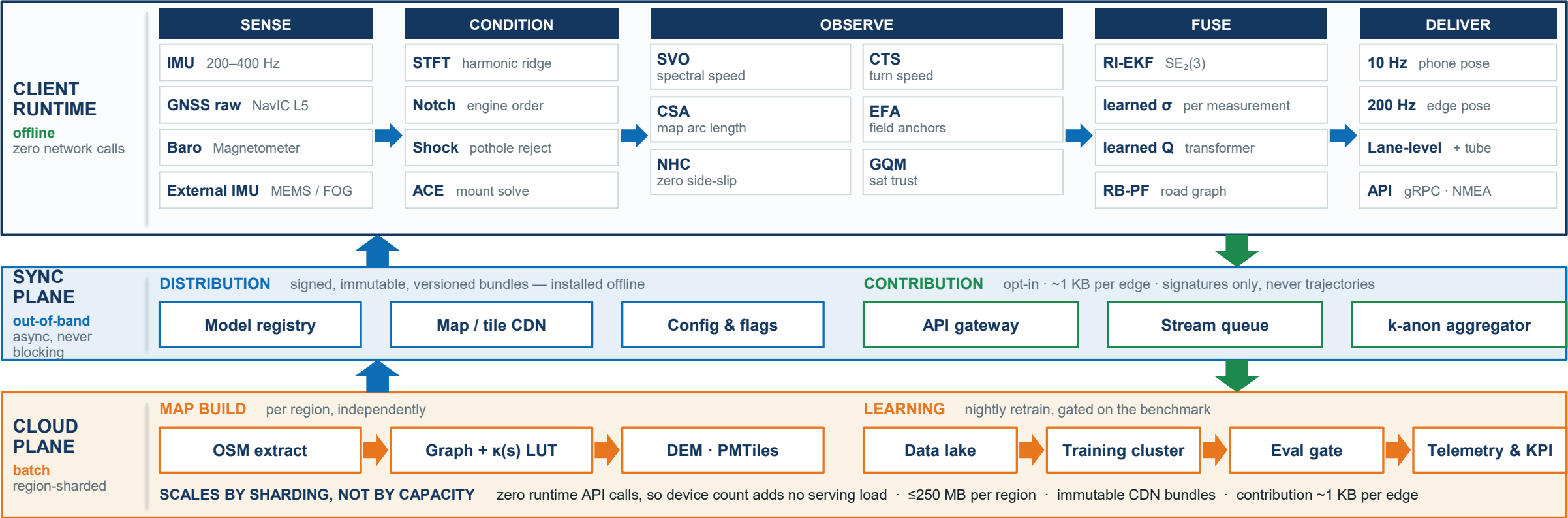
How it addresses the problem



The graph answers one question: after a kilometre with no GPS, how far off are you?

- Pure inertial · 180 m** — acceleration simply added up — tiny errors compound, so you are a block away after one minute.
- Best published phone result · 31 m** — error grows steadily with distance, 3.1 % of every kilometre.
- SETU · 8.5 m** — the line drops each time we recognise something — a bend whose shape matches the map, or a tunnel's magnetic signature. Error is reset instead of piling up.

Methodology and process for implementation



Technologies to be used

Training PyTorch · Lightning · Hydra · MLflow · masked-IMU pretraining	On-device LiteRT + XNNPACK · ExecuTorch · int8 QAT · 3.6 MB total	Engine C++20 · Eigen · $SE_2(3)$ Lie · FlatBuffers · GTest replay	Mobile Kotlin · Compose · MapLibre + PMTiles · SensorDirectChannel	Edge & maps aarch64 · gRPC / NMEA / ROS 2 · osmium · CartoDEM
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FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Data already in hand

IO-VNBD: 5,700 km, phone IMU and CAN bus recorded together

Core physics is closed-form

$v = a_{lat}/\Omega$ and $f = v/2\pi R$ are algebra; only residuals learn

Fits today's phones

3.6 MB int8, 2.7 ms per tick, CPU only — no NPU needed

Sensors already present

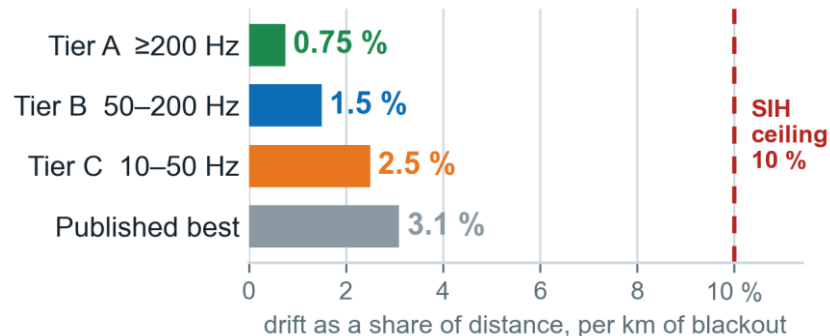
IMU, magnetometer, barometer, raw GNSS with NavIC L5

Maps fit offline

≤250 MB per metro region, no connectivity at runtime

Nothing to buy or fit

no dongle, no encoder, no roadside infrastructure



Every device class clears the ceiling. IO-VNBD's 10 Hz stream is Tier C.

Potential challenges and risks	Severity	Strategies for overcoming these challenges
CSA ambiguous on featureless or badly-mapped roads	High	Landmark saliency scoring · top-k hypotheses in the particle filter · honest σ . Falls back to classical HMM map matching.
Axle harmonics weak — smooth road, quiet EV, soft mount	Medium	Mount-quality gate · summation over 8+ orders · learned σ down-weights it. Falls back to Tier B.
IO-VNBD phone stream is 10 Hz, so cannot validate SVO	Known	Declared as capability tiers. SVO validated on self-collected 400 Hz logs.
No two-wheeler data in IO-VNBD — four cars, no Indian roads	Known	Lean relation imposed as a physics loss, so few samples suffice. Dedicated collection.
Sensor rate throttled below 200 Hz on some devices	Medium	Tier detected at runtime · SensorDirectChannel · verified on three device classes.
Uneven OpenStreetMap geometry on Indian roads	Medium	Covariance scaled by OSM confidence · explicit off-road particle · provider adapter.
Fallback ladder lose CSA → HMM map matching · lose SVO → Tier B · lose both → INS + NHC + ZUPT, already inside the 10 % gate. No single failure leaves us with nothing to show.		



IMPACT AND BENEFITS



Potential impact on the target audience

Delivery and quick-commerce riders

the next turn is announced before the exit, not after it

Logistics and fleet operators

mileage, ETA and proof of delivery survive the gap — no dongle

Ride-hailing drivers and riders

no frozen or teleporting car; pickups on the correct side

Emergency responders

a live position underground, where dispatch loses them today

Two-wheeler riders

India's largest vehicle class, finally with lean-aware physics

Defence and critical logistics

self-contained; degrades gracefully under jamming or spoofing

Benefits of the solution (social, economic, environmental, etc.)

Social

A correct instruction on time means fewer glances at the screen when a rider can least afford one. Emergency crews stay locatable underground. Accurate navigation stops being a feature of expensive cars.

Economic

Zero marginal hardware — it runs on phones riders already own. No dongle per vehicle, no factory INS, no leaky-feeder per tunnel. Fewer missed exits and failed deliveries on the same fleet.

Environmental

A missed exit is wasted kilometres, fuel and emissions; removing the cause removes all three. Entirely on-device, so no roadside infrastructure is built, powered or maintained.

Strategic

Jamming- and spoofing-resilient by construction, since every channel is self-contained. Uses NavIC L5 where Indian devices expose it, runs fully offline, keeps data on the phone.

Where it matters most

Long tunnels and metro underpasses the road is one-dimensional, so the map carries it

Multi-level parking helical ramps are unique signatures; the barometer gives the floor

Dense urban canyons position unusable, but Doppler velocity survives on 3–4 satellites

Forested and valley highways partial GNSS, absorbed by the trust ramp with no mode switch

Pure inertial

180 m

$\frac{1}{2} \cdot b \cdot t^2$ — a block away after one minute

Best published

31 m

3.1 % per km, INS + NHC in a real tunnel

SETU target

8–15 m

0.8–1.5 % per km, anchored to map and field

Horizontal drift over a 1 km blackout at 60 km/h. The SIH ceiling is 10 % — 100 m.

Reach

any Android phone already on the dashboard — no purchase, no fitting, no connectivity. The same engine serves a FOG-grade edge unit at 200 Hz.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

DATASET AND BENCHMARK

IO-VNBD: Inertial & Odometry Benchmark Dataset

Data in Brief 35:106885, 2021 · github.com/onyekpeu/IO-VNBD · [arXiv:2005.01701](https://arxiv.org/abs/2005.01701)

WhONet: Wheel Odometry Neural Network

Eng. Appl. of AI, 2021 · [arXiv:2104.02581](https://arxiv.org/abs/2104.02581) — our privileged upper bound

R-WhONet: recalibration by transfer learning

[arXiv:2209.05877](https://arxiv.org/abs/2209.05877) — evidence of cross-vehicle domain shift

LEARNED INERTIAL ODOMETRY

Deep Learning for Inertial Positioning: A Survey

Chen & Pan · IEEE T-ITS, 2024 · [arXiv:2303.03757](https://arxiv.org/abs/2303.03757)

Inertial Navigation Meets Deep Learning

Cohen & Klein · 2024 · [arXiv:2307.00014](https://arxiv.org/abs/2307.00014)

AI-IMU Dead-Reckoning — invariant EKF

Brossard, Barrau & Bonnabel · IEEE T-IV, 2020 · [arXiv:1904.06064](https://arxiv.org/abs/1904.06064)

TLIO: Tight Learned Inertial Odometry

Liu et al. · IEEE RA-L, 2020 · [arXiv:2007.01867](https://arxiv.org/abs/2007.01867) — learned covariance

OdoNet: untethered speed aiding

[arXiv:2109.03091](https://arxiv.org/abs/2109.03091) · 68 % error reduction over NHC alone

CarSpeedNet: accelerometer-only speed

[arXiv:2401.07468](https://arxiv.org/abs/2401.07468) · $<0.72 \text{ m/s} \approx 4.3 \%$ — the accuracy to beat

LEARNED AND ADAPTIVE FUSION

KalmanNet: NN-aided Kalman filtering

Revach, Shlezinger, van Sloun & Eldar · IEEE TSP, 2022

A-KIT: Adaptive Kalman-Informed Transformer

[arXiv:2401.09987](https://arxiv.org/abs/2401.09987) · github.com/ansfl/A-KIT — learned process noise

PiDR: Physics-Informed Inertial Dead Reckoning

Sahoo & Klein · 2026 · [arXiv:2601.03040](https://arxiv.org/abs/2601.03040)

Differentiable particle filters, semi-supervised

[arXiv:2011.05748](https://arxiv.org/abs/2011.05748) — end-to-end learning with algorithmic priors

MAP MATCHING AND CONSTRAINTS

Hidden Markov Map Matching

Newson & Krumm · ACM SIGSPATIAL, 2009 — the baseline

Heading–length sequence matching

[arXiv:2005.13704](https://arxiv.org/abs/2005.13704) — nearest prior art to CSA; needs a known speed

Map-aided dead reckoning from OBD speed

[arXiv:1611.07910](https://arxiv.org/abs/1611.07910)

Lever-arm accuracy of the non-holonomic constraint

Zhang & Hu · IEEE TVT, 2020

NHC-assisted GNSS/SINS with motion-state CNN

GPS Solutions, 2023 · doi 10.1007/s10291-023-01483-9

SPECTRAL SPEED AND FIELD ANCHORS

Vehicle Speed Tracking Using Chassis Vibrations

Linköping University, 2016 — axle-order tracking, motivates tunnels

Accelerometer-based wheel odometer

Sensors, 2021 · PMC7918720

Magnetic positioning in a long tunnel

Applied Sciences 11(24):11641, 2021

Vehicle positioning in tunnels: a review

Complex & Intelligent Systems, 2025

Barometric floor detection on smartphones

Sensors, 2015 · PMC4431287 — basis for the parking-level anchor

GNSS INTEGRITY AND PLATFORM

Vehicle positioning underground using a smartphone

ISPRS Archives XLVI-3/W1, 2022 — the 3.1 %/km reference

GNSS multipath detection by machine learning

Hsu · IEEE ITSC, 2017

Android GnssMeasurement HAL — raw observables

[source.android.com](https://source.android.com/docs/core/architecture/ndk/api/28/gnss) · pseudorange, Doppler, C/N₀, AGC

u-blox Untethered Dead Reckoning (UDR)

u-blox.com — the closest commercial analogue, on dedicated hardware

A 66-entry annotated bibliography, with a prior-art positioning table for each of the five mechanisms, is maintained alongside the implementation documentation.